



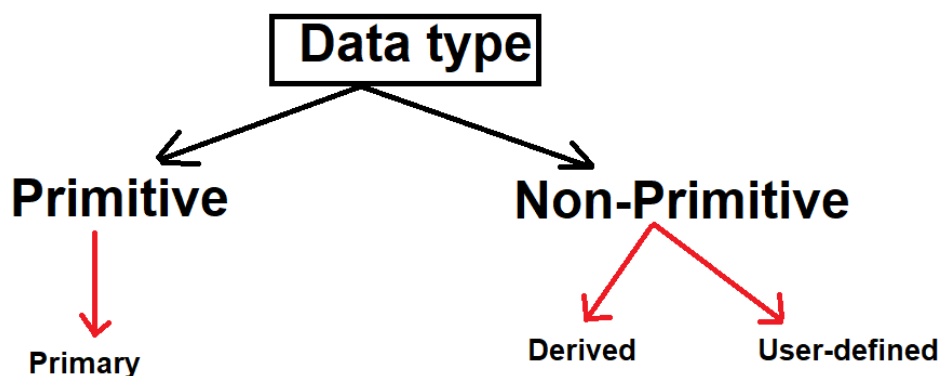
C Programming

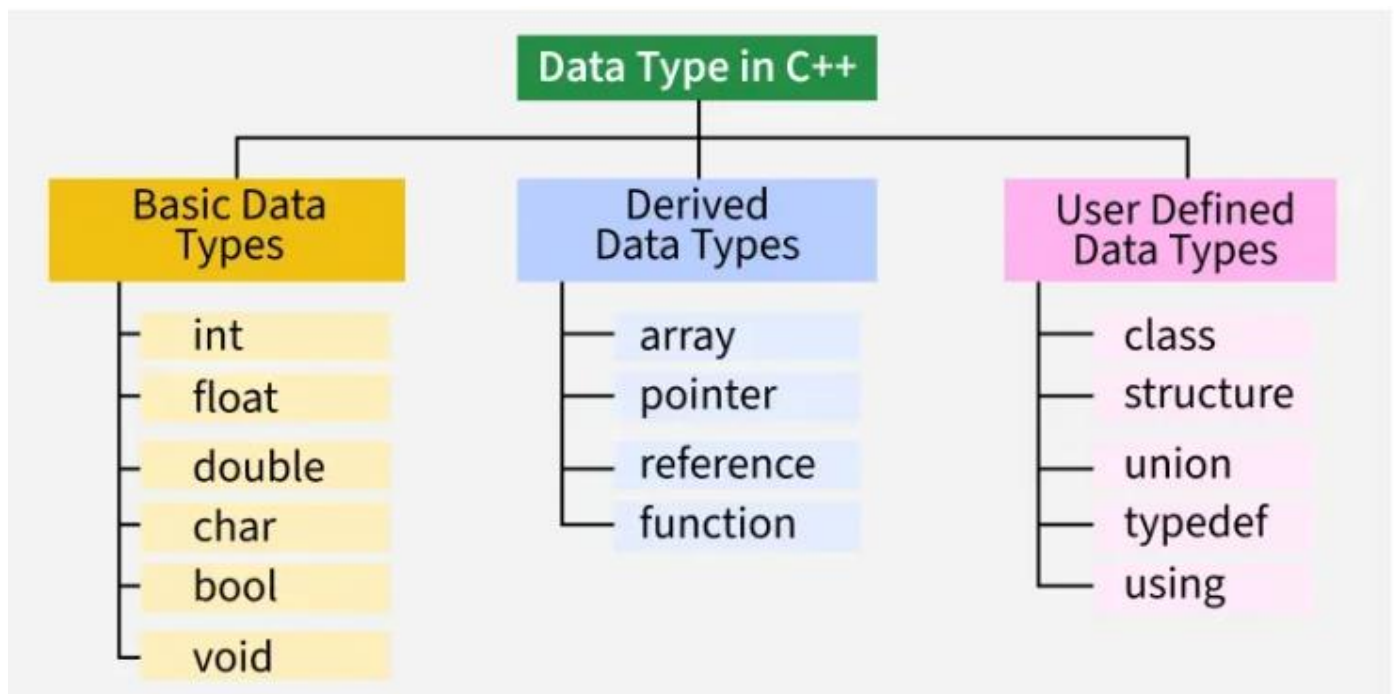
❖ Data Type

- The data type determines the kind of information that may be stored in the variable. Variables are classified according to their data type.
- Whenever a variable is defined in C++, the compiler allocates memory for that variable based on the data type with which it is declared.
- The programmer can select the data type appropriate to the needs of the application. Data types specify the size and types of values to be stored.

C++ supports the following data types:

1. Primary or Built-in or Fundamental/Basic data type
2. Derived data types
3. User-defined data types





❖ Size of Primitive Data type

(64 bit compilers – 2025-2026)

Data Type	Size (byte)	Use	Range
int	4	To store integers numbers	-2147483648 to 2147483647
long	4	For large integers numbers	-2147483648 to 2147483647
long long	8	For very large integers numbers	-9223372036854775808 to 9223372036854775807
float	4	For decimal numbers	$-3.4 * 10^{38}$ to $3.4 * 10^{38}$
double	8	For high precision decimal numbers	$\pm 1.7 * 10^{308}$
long double	8	For extra high precision decimal numbers (some compiler 12/16)	$\pm 1.1 * 10^{4932}$
char	1	To store a single character	'A', 'a', '0',
bool	1	To store true/false values	True/false

Old 16bit /Turbo C++ sizes

Type	Size (bits)	Size (bytes)	Range
char	8	1	-128 to 127
unsigned char	8	1	0 to 255
int	16	2	-2^{15} to $2^{15}-1$
unsigned int	16	2	0 to $2^{16}-1$
short int	8	1	-128 to 127
unsigned short int	8	1	0 to 255
long int	32	4	-2^{31} to $2^{31}-1$
unsigned long int	32	4	0 to $2^{32}-1$
float	32	4	3.4E-38 to 3.4E+38
double	64	8	1.7E-308 to 1.7E+308
long double	80	10	3.4E-4932 to 1.1E+4932

❖ Format Specifiers in C

In C language, format specifiers are used in input and output functions (such as printf() and scanf()) to define the type of data being handled.

Specifier	Type	Description
%d	int	Decimal integer
%c	char	Single character
%s	char[]	String of characters
%f	float	Floating-point number
%lf	double	Double precision floating-point number
%u	unsigned int	Unsigned decimal integer
%x	unsigned int	Hexadecimal integer
%o	unsigned int	Octal integer
%p	void*	Pointer address
%e, %E	float, double	Scientific notation
%g, %G	float, double	Use %e or %f based on value

❖ Let's Start programming in C

Q1. Print "Hello Bihar" in C++ language.

Q2. Print Your name

Q3. Print your name,age,class,address & Gender.

=====HIRA KUMAR=====

